

Learning-Based Control for Stair Ascent in an Insect-Scale Hybrid Quadcopter-Hopper Robot

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Abstract—Robots capable of traversing uneven and discontinuous terrain are essential for real-world deployment, yet achieving robust locomotion remains challenging, particularly for systems with minimal ground contact [11]. This thesis investigates learning-based control for a hybrid quadcopter-hopper robot that combines monopodal hopping with aerial actuation via four wings, where the leg provides impulsive vertical motion and the wings enable stabilization and trajectory control during flight and landing [1]. The primary goal is to enable the robot to reliably ascend a flight of stairs in simulation, a task that serves as a demanding benchmark for terrain-adaptive locomotion due to its reliance on precise timing, orientation control, and robustness to discontinuous contact. Rather than relying exclusively on hand-engineered controllers, this work explores imitation learning and reinforcement learning to synthesize control policies acting directly on the robot’s wing thrust signals, using expert demonstrations from model-based or heuristic controllers and comparing performance against reinforcement learning approaches such as Proximal Policy Optimization (PPO) [2], [8] and model-based baselines. All experiments are conducted in MuJoCo [10], with robustness evaluated across variations in stair geometry and actuator parameters and with particular attention to sim-to-real transfer challenges through domain randomization [6], [9]. The expected outcome is both a functional learning-based controller capable of stair ascent and a comparative analysis of learning-based and model-based approaches for hybrid locomotion, providing insight into how simple, energy-efficient robotic systems can achieve adaptable behavior through learning.

I. INTRODUCTION

Locomotion has long been one of the grand challenges of robotics. In particular, locomotion in unstructured environments remains a central focus in the field today. While wheeled robots are efficient on flat, predictable surfaces, they struggle with obstacles and discontinuous geometry such as stairs. Legged robots offer greater versatility, but achieving robust and adaptable locomotion remains difficult due to intermittent contact, impact dynamics, and sensitivity to modeling error [11].

Among legged systems, monopodal robots present a particularly challenging and informative platform. With only a single point of ground contact, monopodal locomotion requires precise timing, stabilization, and recovery from imperfect landings [1]. Small errors that might be tolerated in multi-legged systems can quickly destabilize monopodal robots, making them a useful testbed for studying dynamic control and robustness.

This thesis focuses on a hybrid quadcopter-hopper robot that combines monopodal hopping with aerial actuation via four wings. In this design, the leg provides impulsive vertical motion, while the wings enable continuous attitude and

trajectory control during flight and landing. Compared to quadrupedal robots, which require coordinated control of many joints and contact points, the hybrid quadcopter-hopper offers a mechanically simpler and more energy-efficient system while still exhibiting rich hybrid dynamics [3]. The separation of roles between leg and wings yields a control problem that is expressive yet relatively low-dimensional, making it well suited for learning-based approaches.

This work builds directly on prior research in the Soft and Micro Robotics Laboratory (SMRL) at MIT, which has developed and experimentally validated insect-scale hybrid winged-hopping robotic platforms and advanced control strategies for insect-scale systems [1], [3]. Whereas previous work in the lab has emphasized carefully engineered and learning-enhanced model-based controllers [4], this thesis explores whether policy learning approaches such as imitation learning and reinforcement learning can provide a complementary pathway toward robustness and terrain adaptation in hybrid locomotion.

A key limitation of many existing locomotion controllers is their reliance on well-characterized terrain. Controllers that perform well on flat ground often degrade significantly when faced with discontinuous terrain such as stairs, where precise body orientation, trajectory control, and robustness to contact uncertainty are required. Biological systems address similar challenges through experience-driven adaptation rather than precise modeling.

Motivated by this gap, this thesis explores learning-based control methods as a means to enable terrain-adaptive locomotion in a hybrid quadcopter-hopper robot. The work centers on a concrete benchmark task: enabling the robot to ascend a short flight of stairs, with extensions to longer staircases explored in simulation.

II. RELATED WORK

A. Hybrid and Monopodal Locomotion

Prior work has demonstrated that combining hopping with aerial actuation can significantly enhance mobility. *An Agile Monopodal Hopping Quadcopter with Synergistic Hybrid Locomotion* shows that wing-generated thrust can stabilize hopping trajectories and enable directional control [1], while *Hybrid Locomotion at the Insect Scale: Combined Flying and Jumping for Enhanced Efficiency and Versatility* draws inspiration from biological systems [3]. These systems establish hybrid winged-hopper robots as viable platforms for

locomotion in environments with sparse or intermittent ground contact.

These approaches typically rely on carefully engineered control strategies, often with scripted leg actuation and feedback-based aerial stabilization. While effective, they generally assume known terrain conditions and do not explicitly address learning-based adaptation to discontinuous environments such as stairs.

B. Model-Based Control

Model predictive control (MPC) has been successfully applied to hopping and legged robots, producing stable behaviors when accurate dynamics models are available [11]. In hybrid winged robots, learning-enhanced MPC frameworks have demonstrated strong robustness and aggressive maneuvering capabilities, as shown in *Aerobic Maneuvers in Insect-Scale Flapping-Wing Aerial Robots via Deep-Learned Robust Tube MPC* [4]. However, MPC approaches can struggle with stair ascent due to non-smooth contact dynamics, uncertainty in contact timing, and sensitivity to modeling error. As a result, MPC is well suited as a baseline or demonstration generator but may be insufficient on its own for robust stair climbing.

C. Learning-Based Locomotion

Reinforcement learning has enabled highly dynamic locomotion behaviors in quadrupedal robots, in some cases outperforming classical control approaches [5]. However, RL policies can be brittle and sensitive to reward design. Imitation learning offers a complementary approach by leveraging expert demonstrations to impose behavioral structure and improve training stability [2], [7]. Despite these advances, learning-based approaches for hybrid monopedal systems, particularly those leveraging aerial actuation for robustness, remain relatively underexplored.

D. Prior Work in the Soft and Micro Robotics Laboratory

Building on prior work in the Soft and Micro Robotics Laboratory that demonstrated robust hybrid locomotion using model-based and learning-enhanced control [3], [4], this thesis explores imitation learning and reinforcement learning as alternative control synthesis approaches for terrain-adaptive hopping. While existing SMRL work focuses on incorporating learning into dynamics modeling and robust control, this project shifts the emphasis toward learning control policies directly from data, enabling adaptation to discontinuous locomotion tasks such as stair ascent.

III. OBJECTIVES

This thesis aims to develop and evaluate learning-based controllers for a hybrid quadcopter-hopper robot, with stair ascent serving as the primary benchmark for terrain-adaptive locomotion.

A. Primary Objective

Enable the robot to reliably ascend a flight of stairs in simulation using learning-based control acting on the four wing thrust signals.

B. Secondary Objectives

Compare imitation learning, reinforcement learning (PPO), and model-based control approaches in terms of robustness and performance.

Evaluate generalization across variations in stair geometry and terrain parameters.

C. Stretch Objective

Explore additional sensory modalities (e.g., terrain perception) to improve generalization if time and resources permit.

IV. METHODOLOGY

A. Task and Simulation Environment

All experiments will be conducted in simulation using MuJoCo. The stair environment is parameterized by step height, depth, friction, and overall staircase length. In simulation, longer staircases may be explored, while physical experiments are constrained to approximately five steps due to space and fabrication considerations.

The robot's motion involves both horizontal and vertical progress, requiring careful coordination of thrust direction, body orientation, and timing. A curriculum-based approach will be used, progressing from flat ground hopping to single-step ascent and eventually to multi-step staircases, a strategy commonly used to stabilize learning in locomotion tasks.

B. Robot Model and Action Space

The robot consists of a single actuated leg and four wings. In line with prior hybrid locomotion work, the leg actuation is scripted to generate consistent hopping impulses, while the learning-based controller outputs four continuous wing thrust commands [1]. This abstraction aligns with existing designs and focuses learning on the aspects of control most critical for robustness: aerial stabilization and trajectory shaping.

C. Control Methodologies

Three primary control approaches will be explored:

1) *Imitation Learning*: Behavioral cloning from expert demonstrations generated using MPC or heuristic controllers [7]. IL provides a strong initialization and improves stability in early training.

2) *Reinforcement Learning*: Proximal Policy Optimization (PPO) [8] is used due to its stability and widespread success in continuous control. Rewards emphasize upward progress, stability, and recovery from imperfect landings.

3) *Model-Based Control*: MPC serves as a baseline and demonstration generator [11]. While MPC may struggle with stair ascent directly, it provides a valuable point of comparison.

Hybrid approaches, such as initializing reinforcement learning from imitation learning or learning residual corrections, may be explored if time permits.

D. Preliminary Results

Preliminary work has focused on establishing the simulation and learning infrastructure required for the proposed experiments. A behavioral cloning pipeline has been implemented in MuJoCo, including demonstration data handling, policy training, and closed-loop rollouts. While this work primarily reproduces existing behaviors rather than producing novel empirical results, it validates the feasibility of the proposed learning framework and reduces project risk.

V. EVALUATION METRICS AND SUCCESS CRITERIA

Performance will be evaluated using both quantitative and qualitative metrics:

- *Success rate*, defined as the percentage of trials in which the robot ascends the staircase without failure.
- *Hop efficiency*, defined as the number of hops required to complete the task.
- *Stability metrics* such as maximum body tilt, angular velocity, and recovery behavior.
- *Robustness tests* to measure performance under perturbed stair geometries and actuator noise.

A successful thesis outcome may include high simulation success rates, a clear comparison across control methods, and a qualitative hardware demonstration, all of which are considered valid contributions.

VI. ANTICIPATED CHALLENGES AND MITIGATION STRATEGIES

The primary challenge is sim-to-real transfer, particularly due to mismatches in thrust dynamics, actuator latency, and contact modeling. This is especially critical for hopping systems, where small errors can accumulate across repeated impacts.

Mitigation strategies include domain randomization of actuator response, friction, and contact parameters; action smoothing and rate limits; and training across varied stair geometries [6], [9]. Emphasis is placed on robustness rather than optimality, prioritizing policies that tolerate modeling error over those that exploit simulator-specific dynamics.

Additional challenges include policy brittleness and overfitting to simulator artifacts. These will be addressed through evaluation on held-out terrain configurations, incorporation of imitation-based priors [2], and conservative regularization emphasizing robustness over optimality.

VII. PROPOSED TIMELINE

The following is a proposed timeline for the thesis.

Month	Milestone
January	Finalize stair task specification
	Environment setup
	Baseline controller refinement
February	Generate expert demonstrations
	Train imitation learning policies
March	Train PPO policies
	Compare IL and RL approaches
April	Robustness and sim-to-real focused experiments
May	Consolidate results
	Finalize thesis writing

VIII. BROADER IMPACT AND SIGNIFICANCE

This work contributes to a growing body of research emphasizing minimalist robotic design. Rather than increasing mechanical complexity, the hybrid quadcopter–hopper demonstrates how learning-based control can enable adaptable behavior in simple, energy-efficient systems. Insights from this project may inform future work in small-scale robotics, bio-inspired locomotion, and hybrid aerial–ground systems.

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